The Bid Optimization Problem

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TII - Technology Innovation Institute

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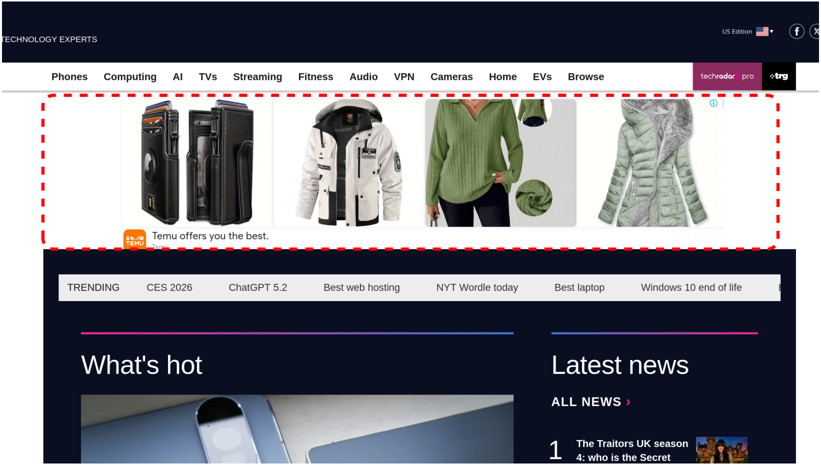
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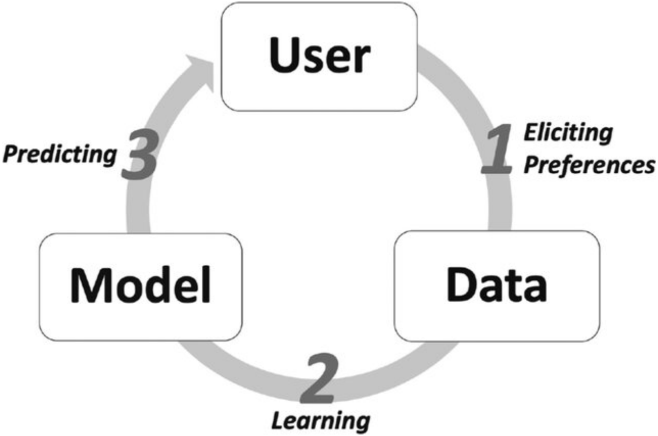
Standard RecSys flow



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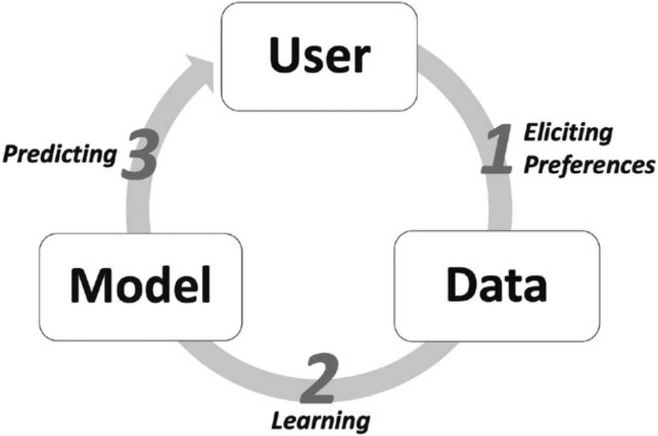
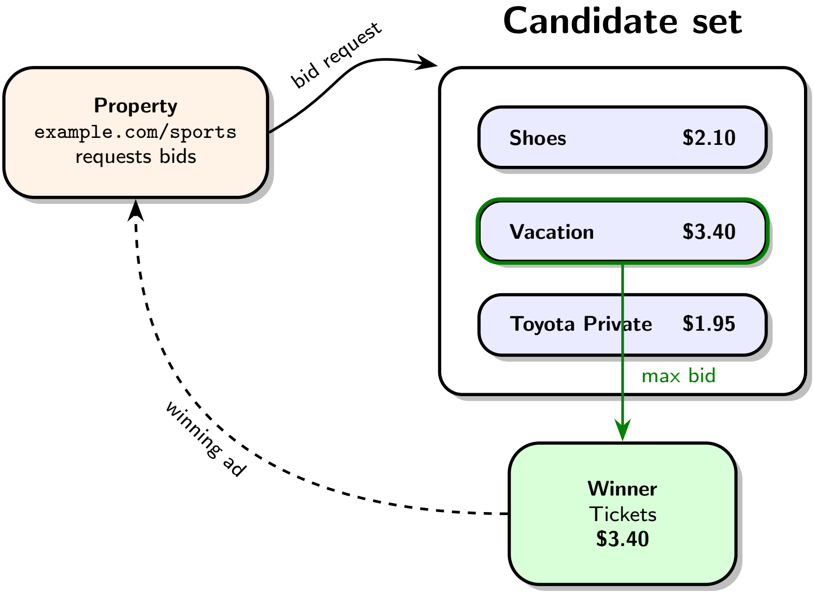
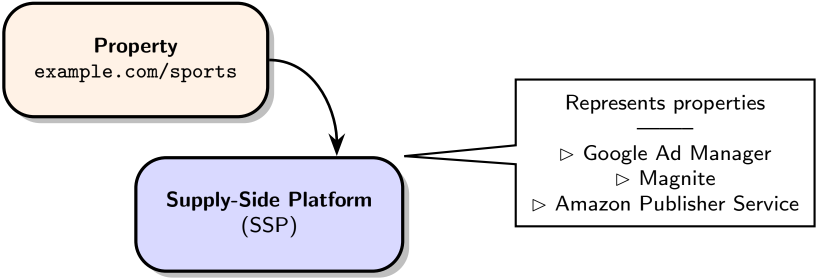
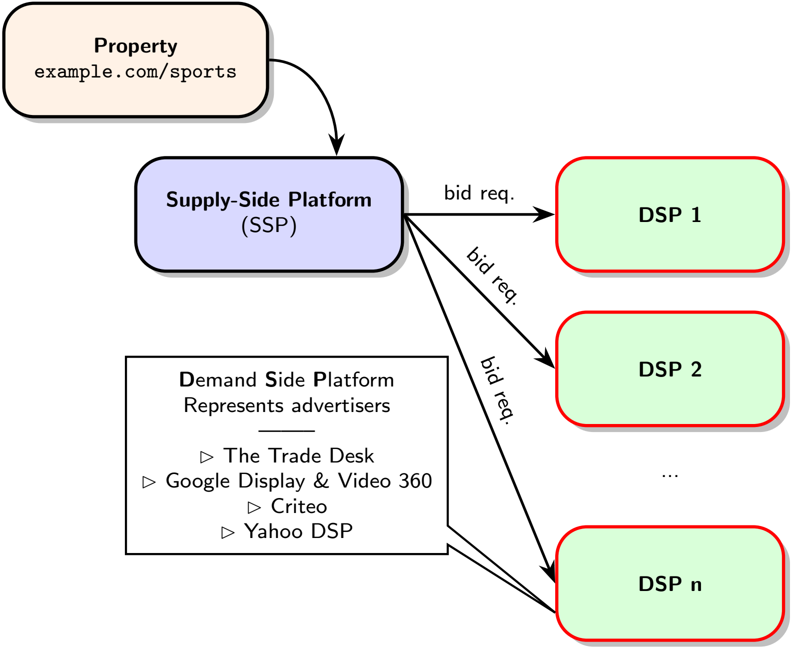


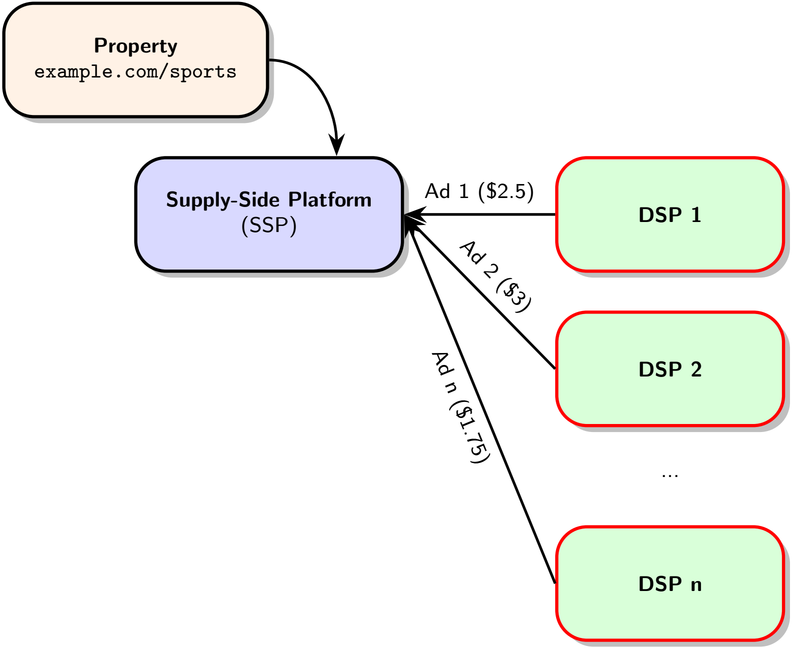
In ads - recommendation is not just "predicting".

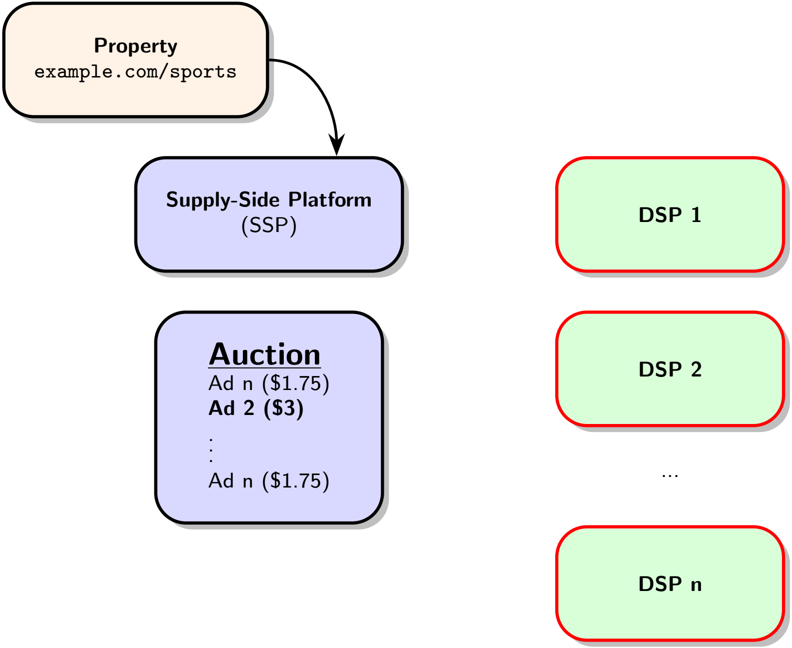


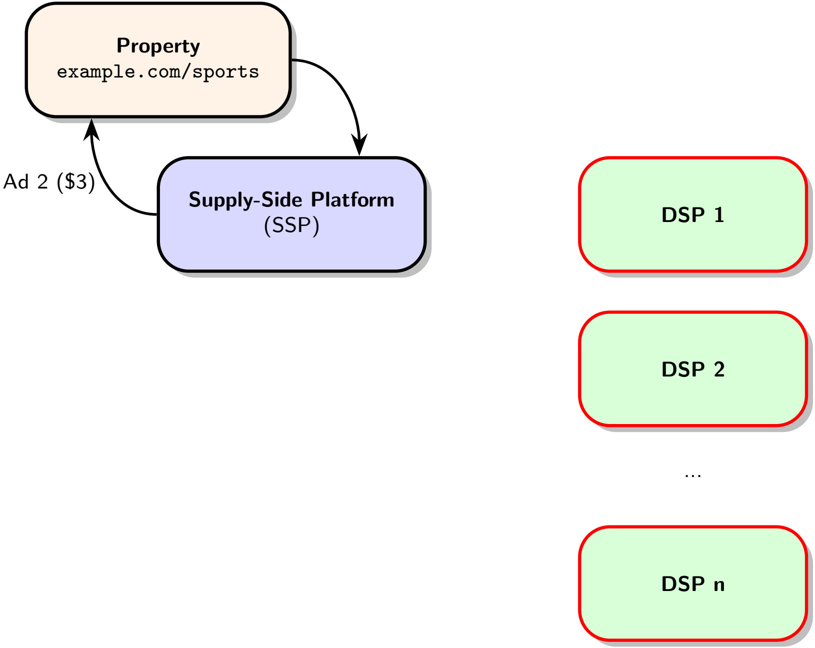
Candidate set
bid request
Property
example.com/sports
requests bids
Shoes $2.10
Vacation $3.40
Toyota Private $1.95
max bid
winning ad
Winner
Tickets
$3.40

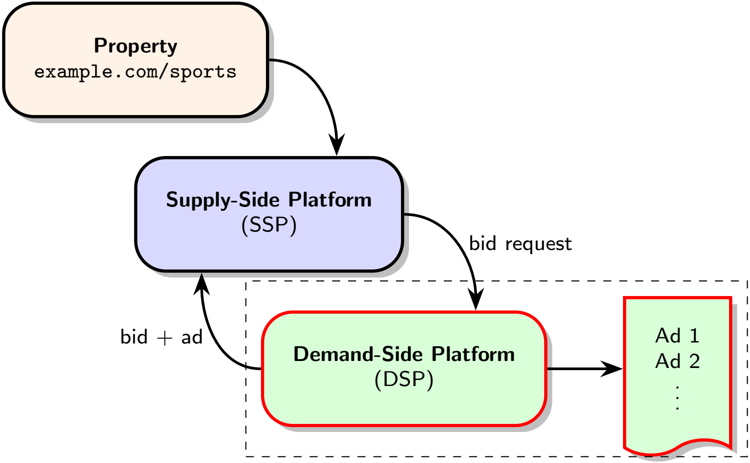

Property
example.com/sports
Supply-Side Platform
(SSP)
Represents properties
▷ Google Ad Manager
▷ Magnite
▷ Amazon Publisher Service

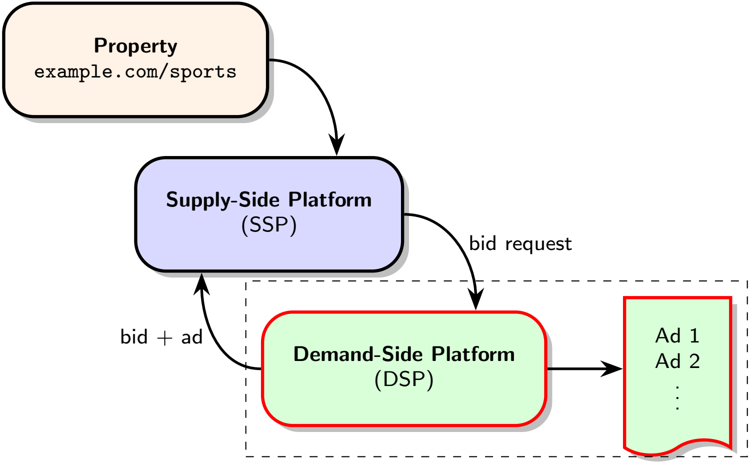

Property
example.com/sports
Supply-Side Platform
(SSP)
bid req.
bid req.
bid req.
DSP 1
DSP 2
...
DSP n
Demand Side Platform
Represents advertisers
▷ The Trade Desk
▷ Google Display & Video 360
▷ Criteo
▷ Yahoo DSP


Property
example.com/sports
Supply-Side Platform
(SSP)
Ad 1 ($2.5)
Ad 2 ($3)
Ad n ($1.75)
DSP 1
DSP 2
...
DSP n


Property
example.com/sports
Supply-Side Platform
(SSP)
Auction
Ad n ($1.75)
Ad 2 ($3)
Ad n ($1.75)
DSP 1
DSP 2
...
DSP n


Property
example.com/sports
Ad 2 ($3)
Supply-Side Platform
(SSP)
DSP 1
DSP 2
...
DSP n


Property
example.com/sports
Supply-Side Platform
(SSP)
bid request
bid + ad
Demand-Side Platform
(DSP)
Ad 1
Ad 2
⋮



Objective - in *a few milliseconds*, select an **ad** coupled with a **bid**

Primary sources of revenue

- Transaction fees (take rate)
- Premium services
- Subscription (annual / quarterly fee)

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Business goal - balance short term revenue and long term satisfaction.

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  - max. exposure
  - avg cost per purchase
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  - b is the bid
  - $\boldsymbol{x} = (\boldsymbol{x}_{\mathrm{ad}}, \boldsymbol{x}_{\mathrm{ctx}})$ is a vector of ad, and context+user features

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    $\boldsymbol{x}_{\mathrm{ad}}$ ex: product price, product category, ...
    $\boldsymbol{x}_{\mathrm{ctx}}$ ex: site domain, time of day, user device, user OS, ...

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The optimal bid for a given ad, context, and user:

$$b^*(\boldsymbol{x}) = \operatorname*{argmax}_b u(b, \boldsymbol{x})$$

Utility example - avg cost per purchase

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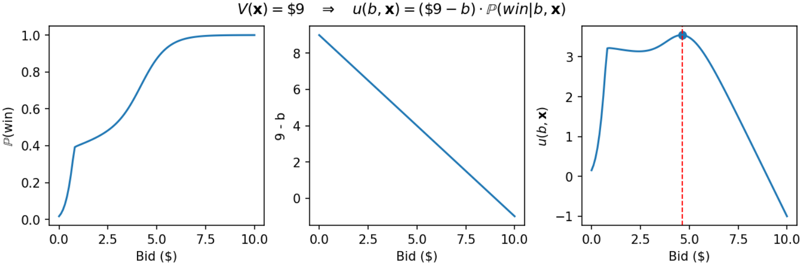

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Machine learned

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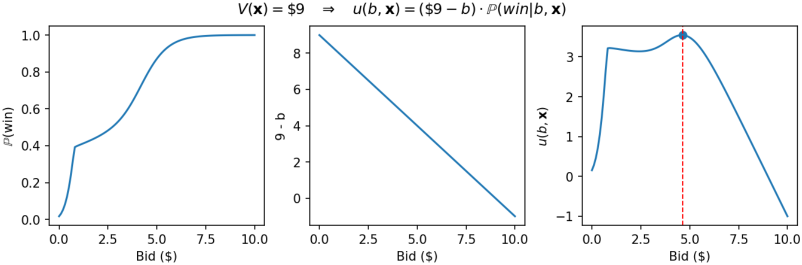

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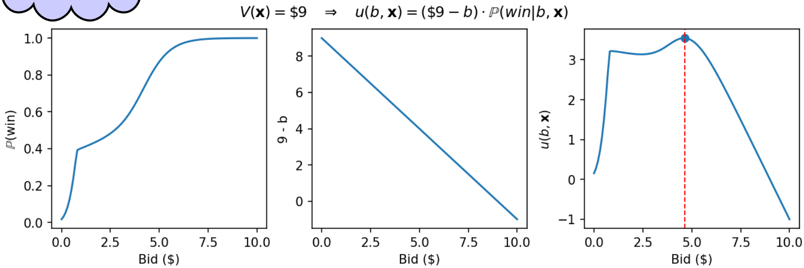
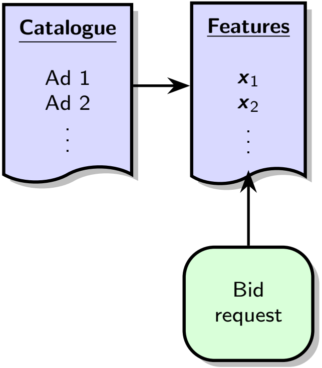


DSP ad choice - ideal vs practice

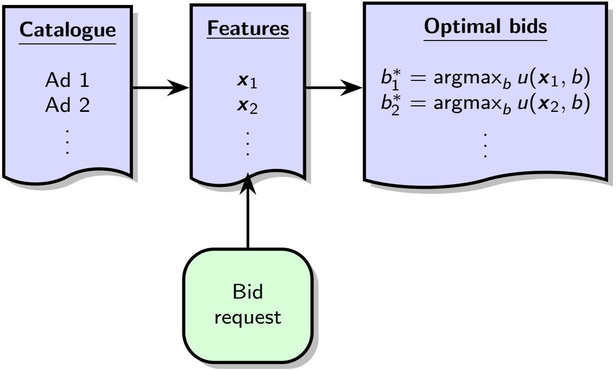
Catalogue

Ad 1
Ad 2
$\vdots$

DSP ad choice - ideal vs practice



DSP ad choice - ideal vs practice



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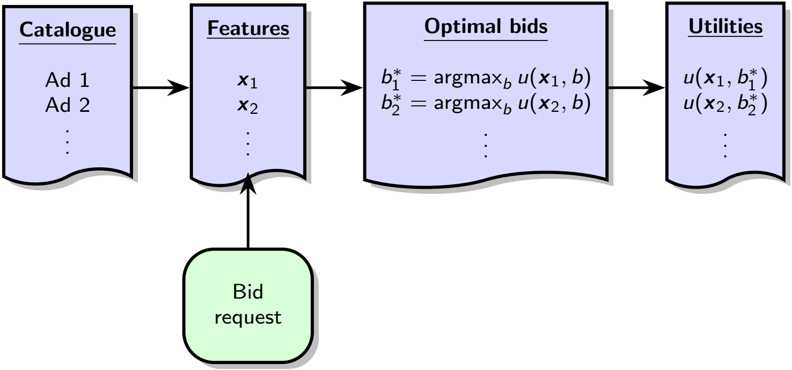

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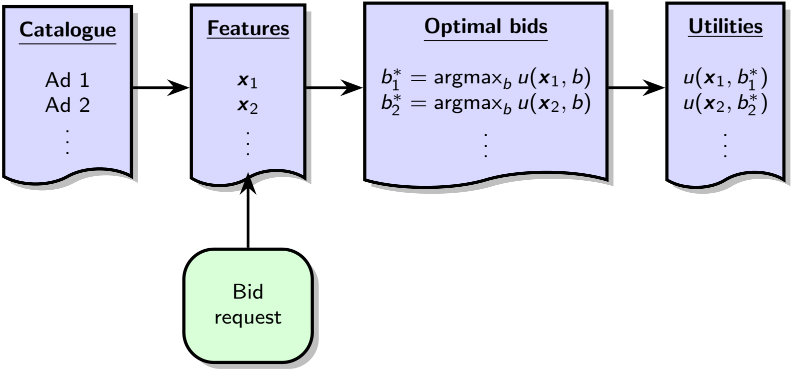

Highest utility ad(s) is chosen

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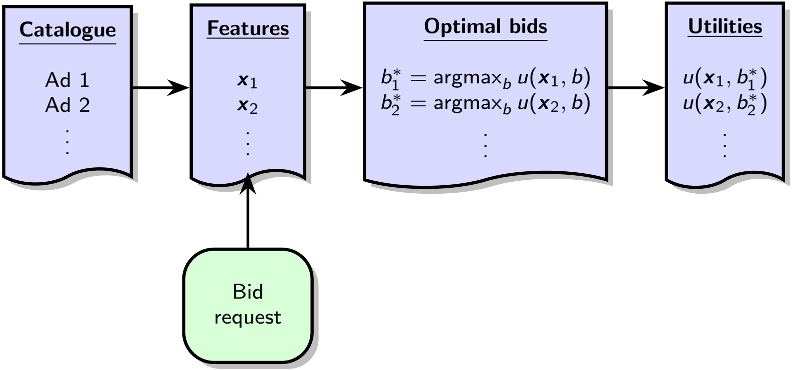

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Now always practical. Phased ranking is typically used.

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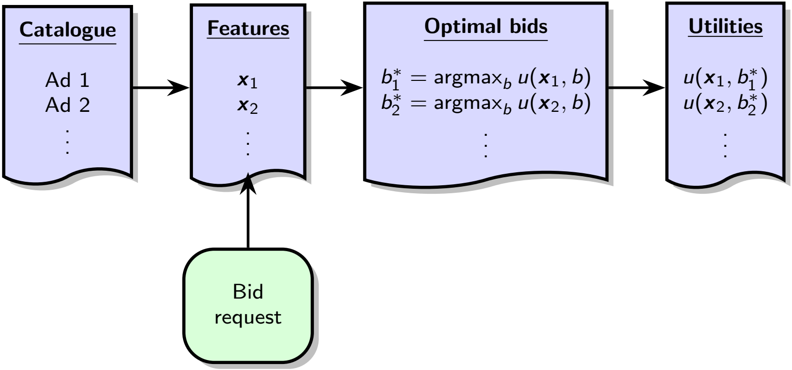

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Example: phase 1 - rank by value, phase 2 - compute bids and rank by utility.

Challenges

Model types

- Value prediction models: click probability / purchase probability / ...
- Win probability prediction models

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Main challenge - balance

- Training speed
- Prediction / inference speed
- Model performance

Some typical models

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Factorization machines [Rendle et. al. 2016]

$$\Phi_{\mathrm{FM}}(\boldsymbol{x}) = a + \langle \boldsymbol{w}, \boldsymbol{x} \rangle + \sum_{i=1}^{n} \sum_{j=i+1}^{n} \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle x_i x_j$$

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$$= a + \langle \boldsymbol{w}, \boldsymbol{x} \rangle + \frac{1}{2} \left\| \sum_{i=1}^{n} x_i \boldsymbol{v}_i \right\|^2 - \frac{1}{2} \sum_{i=1}^{n} \| x_i \boldsymbol{v}_i \|^2$$

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Time linear in the dimension of $\boldsymbol{x}$!

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Field-weighted factorization machines [Pan et. al. 2018]

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MANY more (DCN, DCNv2, Deep&Wide, DeepFM, ...)

Win rate prediction models

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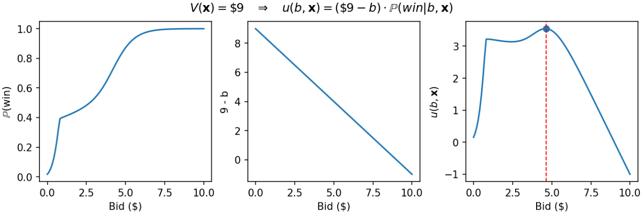

$\mathbb{P}(\mathrm{win}|\boldsymbol{x}, b)$ HAS to be a fast-to-compute function of b!

Bid Shading by Win-Rate Estimation and Surplus Maximization (2020, Pan et. al.):

$$\mathbb{P}(\mathrm{win}|\boldsymbol{x}, b) = \sigma(\Phi(\boldsymbol{x}) + c \cdot \ln(b)),$$

where $\sigma(x)$ is the sigmoid.

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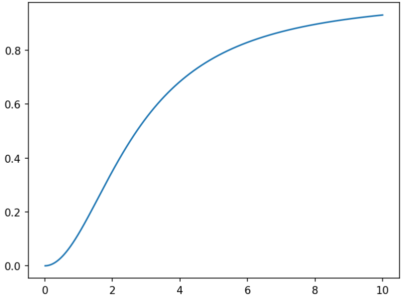

MEOW: A space-efficient nonparametric bid shading algorithm (2021, Zhang et. al.)

Instead of $\sigma(\Phi(\boldsymbol{x}) + c \cdot \ln(b))$, learn a table:

bid \ valuation	v_1	v_2	$\cdots$	v_j	$\cdots$	v_M
b_1				$S_{1,j}$		
b_2				$S_{2,j}$		
$\vdots$				$\vdots$		
b_i				$S_{i,j}$		
$\vdots$				$\vdots$		
b_K				$S_{K,j}$		

Recent trends

Use generative RL-trained models for auto-bidding.

- "Generative Auto-bidding via Conditional Diffusion Modeling" (2024, Guo et. al.)
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Limited usefulness - compute and data heavy.

Questions?